

Aidan Ryder

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ABOUT ME

Design Engineering student at Imperial College London and Kitchener Scholar, with strengths in systems thinking, teamwork and prototyping. I am particularly drawn to complex, open-ended problems where I can research unfamiliar areas, connect ideas across disciplines and develop practical solutions with real-world impact. Seeking a 6-month industrial placement as part of my degree.

EDUCATION

MEng Design Engineering

2024 – 2028

Imperial College London

London

- MODULE CONTENT: Human Centred Design, Industrial Design (DFMA), UI & UX Design, Mechatronic Design, Sustainable Design, System Thinking, FEA & CFD (Ansys), Web Application Design.

SKILLS

Software

Figma, Fusion 360, Ansys Fluent, Ansys Static Structural, Ansys Modal, Overleaf, MATLAB, Visual Studios Code, Notion

Design

CAD, Circuit Design, Mechanical Design, Systems Design, Business Modelling, Branding, Human Centred Design

Soft Skills

- Teamwork, Communication Skills, Time Management (projects range from 1-2 terms 3-6 months), Presentation Skills, the ability to communicate through stylised & technical reports.

PROJECTS

Axis Pro

- Group: Developed a product for an underserved user group and iteratively refined the design around user needs. The final electromechanical product provided controlled vertical translation and rotational movement for precision painting, alongside packaging and overall branding. We also produced a video showcasing the final product and documenting the design process.
- As the designated CTO, my main focus was the electromechanical and Design for Manufacture and Assembly (DFMA) aspects of the project. I focused on reducing manufacturing time and cost through industrial design principles, mechanical simplification and consideration of appropriate manufacturing methods. This project also allowed me to develop my leadership skills as I often led meetings and set tasks for other team members which gave me experience helping manage the overall direction of the project.

Airline Seat System Design

- Group: As a group we dismantled a commercial airline chair and talked to industry leaders about what an airline chair needs, the system behind building & selling the chair as a product and how this system might be changed or adjusted/improved upon to create a more economically, environmentally and socially sustainable product. This side of the project also helped me develop my teamwork skills, particularly through debating different ideas and reaching decisions in a productive way.
- Individual: This section was about condensing our knowledge into designing a new system and chair that would fit that system and discussing how it might be implemented into the current market.

Prosthetic Leg

- Individual: Designed and modelled a prosthetic running leg in Fusion 360 to meet predefined performance criteria, including a fatigue life of at least 1 million loading cycles and a fundamental frequency above 50 Hz. ANSYS modal and static structural analyses were conducted, with mesh refinement used to verify result convergence and improve confidence in the simulation. Results were used to identify structural weaknesses and inform geometry changes for an improved second design.
- Post the analysis of the second design, there was further analysis into the performance difference between using a CFRP or aluminium alloy for the blade material.

Gizmo

- Group: In a team of two, designed an interactive mechanical toy combining multiple user inputs, including levers, buttons and ToF sensors, with both light-based and mechanical outputs. Features included mechanisms such as a Scotch yoke paired with transparent components to create the illusion of “magic” as users interacted with the device.
- Integrated the individual subsystems so they could function both independently and together as a single immersive experience.
- The project was selected for display at the Imperial College Open Day.

Rhythm Rush

- Group: Gathered user data through interviews, observations and structured experiments to understand human behaviour in urban spaces and identify opportunities to increase urban play. Used these insights to develop a human-centred concept for situations where interaction with strangers is socially acceptable, resulting in a body-tracking dance game designed to encourage spontaneous social interaction.

WORK HISTORY

Paddington Robotics	06/2026 London
Assembled supermarket robots and discussed DFA choices and pointed out possible improvements. The robot assembly was meant to take 3-4 weeks in teams of two, but through efficiency we cut it down to 1 week with a team of two. This work experience taught me a lot about the possible benefits of asymmetry in design.	
McDonalds	08/2023 – 07/2024 Wirral
• Working multiple roles under high time pressure. By the evenings I was often working 2-3 roles simultaneously. • I taught newer employees how to use the grill as efficiently as possible while cleaning it at high temperatures on the go.	

INTERESTS

Sports	Reading	General
Competed in half marathons and enjoy mountain running, casual climbing and general gym/training.	Reading history books, classics, philosophy and fiction.	Riding motor bikes and playing chess